

Project Proposal

Group 30

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1 Overview

This project focuses on training a machine learning model to accurately segment brain tumors from MRI scans using a 2D U-Net convolutional neural network. Brain tumor classification involves identifying tumor regions within an MRI image.

Automated systems have the potential to support doctors by providing an initial segmentation that can be reviewed by the doctor, helping reduce the time for manual segmentation. Further, automated systems can in many cases act as a proof-reader, providing an additional safeguard towards any false-negatives.

Brain tumor classification is a challenging machine learning problem since tumors can vary significantly in appearance between subjects. Further, tumor tissue may also be similar visually to surrounding healthy tissue, or may only take up a small region of the MRI image.

2 Data Source(s) and Plan for Data Collection

2.1 yashika TODO

open data points example in itk snap and confirm slice indices

2.2 Primary Data Source

The primary data source is the **RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) 2021 Challenge dataset** (Task 1: segmentation). It is the standard multi-institutional, expert-annotated multi-modal MRI benchmark for glioma segmentation. Each subject provides four co-registered MRI volumes (T1, T1CE, T2, T2-FLAIR) and an expert segmentation mask, which matches our design of stacking the four modalities as input channels to a 2D U-Net and predicting pixel-wise tumor sub-region labels.

2.3 Official Sources and Access

- **Primary challenge platform (Synapse / Sage Bionetworks):**

<https://www.synapse.org/#!/Synapse:syn51514105> (training data under syn51514105 or successor links). The project uses the BraTS adult glioma MRI dataset, originally associated with the BraTS 2020/2021 challenges. The original BraTS 2021 challenge is now closed and its dataset is no longer directly hosted through the original challenge page. According to the BraTS organizers,

the BraTS-GLI dataset made available through the BraTS 2023 challenge contains the same BraTS 2020/2021 data, with the enhancing tumor (ET) label changed from 4 to 3.

- **The Cancer Imaging Archive (TCIA) mirror / original contributing collections:** DOI <https://doi.org/10.7937/jc8x-9874> (RSNA-ASNR-MICCAI-BraTS-2021). This hosts the processed NIfTI volumes (and some DICOM) for approximately 1480 subjects of the challenge cohort under a CC-BY 4.0 license. Underlying raw DICOM from contributing collections (TCGA-GBM, TCGA-LGG, IvyGAP, CPTAC-GBM, UPENN-GBM, ACRIN-FMISO-Brain, UCSF-PDGM, etc.) are available via TCIA's NBIA Data Retriever under the respective collection licenses (mostly open or controlled-access with DUA).
- **Kaggle** (convenient for download, but secondary): example <https://www.kaggle.com/datasets/dschettler8845/brats-2021-task1> Related uploads such as [awsaf49/brats2020-training-data](https://www.kaggle.com/datasets/awsaf49/brats2020-training-data) for earlier years are available.

2.4 Terms of Service, Licensing and Accessing the Data

- **No web scraping is performed.** The data is a formally published research dataset obtained through official portals (Synapse client, TCIA download tools) or the Kaggle mirror. There is no live API polling, so rate limiting is not a concern. Large downloads use the authenticated bulk-download tools provided by these platforms.
- **License:** released for research use with attribution (CC-BY 4.0 on TCIA). We will use the data for non-commercial academic purposes only and will not attempt to re-identify subjects. All data was de-identified by the organizers before release.
- **Required acknowledgement:** we will include the acknowledgement statement requested by the organizers, referencing the RSNA-ASNR-MICCAI BraTS Challenge and Synapse ID syn25829067 and cite the papers listed in the Citations subsection.

2.5 How the Data Were Collected, Annotated and Pre-processed

Acquisition: Retrospective multi-institutional routine clinical pre-operative multiparametric MRI (mpMRI) of adult gliomas (WHO grades II–IV, predominantly glioblastoma / high-grade and lower-grade gliomas).

Scanners and protocols vary across approximately 15 institutions (United States, Europe, India, etc.), producing realistic heterogeneity in field strength (1.5T / 3T), sequence parameters and image quality.

Modalities: native T1, post-contrast T1-weighted (T1Gd / T1CE), T2-weighted and T2-FLAIR. Each subject contributes four co-registered 3-D volumes.

Annotation protocol: Labels were produced by manual delineation by 1 - 4 trained raters following a standardized BraTS protocol, then reviewed and approved by experienced neuroradiologists. **We do not label any data ourselves**, so no per-instance annotation time applies. (Manual voxel-level delineation of a single 3D glioma volume is a labor-intensive expert task, which is precisely why using a pre-annotated benchmark is appropriate.) The mask labels are:

Label	Region	Abbrev.
0	Background / non-tumor	—
1	Necrotic and non-enhancing tumor core	NCR/NET
2	Peritumoral edema / invaded tissue	ED
4	GD-enhancing tumor	ET

Standard evaluation regions are Whole Tumor = $\{1, 2, 4\}$, Tumor Core = $\{1, 4\}$, and Enhancing Tumor = $\{4\}$. Label 3 does not exist in BraTS masks. Before computing the loss we will remap $4 \rightarrow 3$, giving a 4-class problem with labels $\{0, 1, 2, 3\}$.

Preprocessing applied by the organizers: DICOM to NIfTI conversion (with PHI removed), rigid co-registration of the four modalities, registration to the SRI24 anatomical template, resampling to 1 mm^3 isotropic resolution, and skull-stripping. Volumes are $240 \times 240 \times 155$ voxels stored as `.nii.gz`.

Preprocessing performed by us: Intensities have no universal scale in MRI, so we will apply per-volume, per-modality z-score normalization computed over non-zero (brain) voxels only, optionally with percentile clipping. We will then stack the four modalities as channels, extract axial 2D slices, remap the labels as above, and apply training-time augmentation (flips, small rotations, elastic deformation, intensity shifts) applied identically across channels and mask.

2.6 Available Metadata

Each case is identified by a subject ID (BraTS2021_XXXXX). The release also provides, for subsets of cases, a mapping to original TCIA/TCGA identifiers and MGMT promoter methylation status (used in the radiogenomic Task 2). Metadata is not used as a model input in this project the model input is the four imaging modalities only.

2.7 Dataset Size and Use of the Data

- **3D volumes:** 1251 publicly labeled training subjects. Each contributes four modality volumes and one mask. A further 219 validation cases are released with hidden labels and cannot be used for local evaluation, so all train/validation/test splits are drawn from the 1251 labeled subjects.
- **2D slices:** each subject has 155 axial slices, giving roughly 194,000 slices in total. After discarding empty (all-zero) slices and subsampling slices without tumor, we expect on the order of tens of thousands of multi-channel training slices. The exact count will be computed after download. This far exceeds the 1000-data-point minimum.
- **Entire dataset vs subset:** we plan to use the *entire* labeled cohort. Tumor pixels are a small minority of each slice, so every tumor-containing slice is valuable and inter-subject variability in tumor appearance and scanner characteristics supports generalization. If memory or compute limits arise during prototyping, a random subject-level subset (30–50%) may be used temporarily, with final models trained on the full set.
- **Splitting:** the train/validation/test split (approx. 70/15/15) is done *by subject*, never by slice, to prevent leakage between adjacent slices of the same patient.

Why the entire labeled training cohort rather than a small subset. The project goal is accurate segmentation under realistic appearance variation. Using the full 1251-subject set (and the derived 2-D slices) maximizes statistical power, captures the documented inter-subject heterogeneity of tumor morphology and intensity and permits rigorous evaluation of the 2-D U-Net against published BraTS benchmarks. Computational resources permitting, the complete set will be used. If memory constraints arise during early prototyping, a random but stratified 30 - 50 % subject subset can be employed temporarily, with final models re-trained on the full set.

2.8 Example Data Points

Each “data point” for the 2-D U-Net is a 4-channel 240×240 (or cropped/resized) axial slice together with its corresponding integer label map.

1. **Subject BraTS2021_00000, axial slice $\approx z = 78$** Input: stacked [T1, T1CE, T2, FLAIR] intensities (skull-stripped, co-registered). Label map: predominantly background (0) with a central necrotic core (1), surrounding enhancing rim (4) and peripheral edema (2). Typical whole-tumor Dice target region occupies $\sim 5 - 15\%$ of the slice.
2. **Subject BraTS2021_00123, axial slice $\approx z = 95$** Input: same four modalities. Label map: larger infiltrative edema (2) with smaller non-enhancing core (1) and sparse enhancing foci (4). Demonstrates the visual similarity of tumor tissue to surrounding healthy white matter that makes the task challenging.
3. **Subject BraTS2021_00456, axial slice $\approx z = 62$** Input: four modalities. Label map: almost pure enhancing tumor (4) with minimal necrosis, illustrating inter-subject morphological variability. Background class still dominates the pixel count, motivating Dice / focal / class-weighted losses.

2.9 Citations

The following works are cited for use of this dataset:

1. U. Baid et al., “The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification,” arXiv:2107.02314, 2021.
2. B. H. Menze et al., “The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS),” *IEEE Trans. Med. Imaging*, 34(10):1993–2024, 2015.
3. S. Bakas et al., “Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features,” *Scientific Data*, 4:170117, 2017.
4. Kaggle mirror: <https://www.kaggle.com/datasets/dschettler8845/brats-2021-task1>.

3 Solution

U-Net is a convolutional neural network (CNN) architecture designed for image segmentation tasks. This architecture consists of an encoder and decoder. The encoder repeatedly applies convolutions and downsamples the image. It learns increasingly abstract features. The decoder upsamples these features back up to the original resolution of the input. Its job is to figure out the exact location of features. In the context of our project, encoded early layers will detect edges and features, while deeper layers can learn patterns associated with tumors. [?]

Each MRI scan consists of four images: T1, contrast-enhanced T1 (T1CE), T2, and T2-FLAIR. These images will be stacked as separate input channels, allowing the model to use information from each image concurrently in its prediction.

The target labels will consist of the already labeled segmentation masks provided with the dataset. Therefore, as the labels are pixel level, the model will also perform pixel-wise classification, assigning each pixel in the U-Net slice to either the background or one of the annotated tumor regions.

The model will be built based off a U-Net architecture, which is a tried and proven workflow designed specifically for biomedical image classification tasks. PyTorch will be used for implementation and training/validation.